

Source identification in the presence of an external velocity field using dynamic optimal transport

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Abstract

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1 Problem statement

Definition 1.1. *We consider the variational dynamic problem*

$$\inf_{(f_t; J_t, \rho_t; \rho_0, \rho_1) \in \text{CE}} \int_0^1 \int_{\Omega} \frac{|J_t|^2}{2\rho_t} dxdt + \alpha_f \int_0^1 \int_{\Omega} \frac{|f_t|^2}{2\rho_t} dxdt + \frac{\alpha_{\text{inv}}}{2} \int_0^1 \int_{\Gamma} |\rho(t) - \rho_{\text{data}}|^2 dxdt, \quad (\text{BB2})$$

for given curve $\Gamma \subset \Omega \times (0, 1)$, where CE denotes the set of all (ρ_t, J_t) being solution to

$$\begin{aligned} \frac{\partial \rho_t}{\partial t} + \nabla \cdot (J_t + \rho_t v_t) &= f_t, \quad \text{on } \Omega \times (0, 1) \\ (J_t + \rho_t v_t) \cdot n &= 0 \quad \partial\Omega \times (0, 1). \end{aligned} \quad (1)$$

Depending on the specific situation, initial- and final data may or may not be imposed:

$$\rho_0 = \rho^0, \quad \text{and} \quad \rho_1 = \rho^1.$$

Equation (1) automatically enforces the constraint

$$\int_{\Omega} d\rho_1 - \int_{\Omega} d\rho_0 + \int_0^1 \int_{\Omega} f_t dxdt = 0.$$

However, in our numerical treatment, we may impose it explicitly.

2 Numerics

In this Section we present a primal-dual optimization algorithm solving the dynamic optimal transport introduced above

2.1 Augmented Lagrangian formulation

Following [?], we can rewrite the minimization problem (BB2) as an unconstrained one by employing a suitable Lagrangian. Let

$$\begin{aligned} \mathcal{G}(\phi_t) &:= - \int_{\Omega} [\phi_1 d\rho_1 - \phi_0 d\rho_0], \\ \langle \phi_t, \rho_t \rangle_{\Omega} &:= \int_0^1 \int_{\Omega} \phi_t d\rho_t dt \end{aligned}$$

where $\phi_t : [0, 1] \times \Omega \rightarrow \mathbb{R}$ denotes a smooth test function. Defining ϕ_t as test-functions in (1) allows to rewrite (BB2) as the saddle-point problem

$$\begin{aligned} \inf_{(f_t, \rho_t, J_t) \in \mathcal{D}_{\text{adm}}} \sup_{\phi_t, \lambda_t} & \int_0^1 \int_{\Omega} \Psi(\rho_t, J_t) dt + \alpha_f \int_0^1 \int_{\Omega} \Psi(\rho_t, f_t) dx dt + \frac{\alpha_{\text{inv}}}{2} \int_0^1 \int_{\Gamma} |\rho(t) - \rho_{\text{data}}|^2 dx dt \\ & - \mathcal{G}(\phi_t) - \langle (\partial_t \phi_t, \nabla \phi_t, \phi_t), (\rho_t, J_t + \rho_t v_t, f_t) \rangle_{\Omega} \\ & + \lambda_t \left(\int_{\Omega} d\rho_1 - \int_{\Omega} d\rho_0 + \int_0^1 \int_{\Omega} f_t dx dt \right), \end{aligned}$$

with

$$\mathcal{D}_{\text{adm}} = M_{[0,1]}(\Omega) \times M_{[0,1]}(\Omega) \times M_{[0,1]}(\Omega, \mathbb{R}^d) \quad (2)$$

and

$$\Psi(u, v) := \begin{cases} \frac{|v|^2}{2u} & \text{if } u > 0, \\ 0 & \text{if } u = 0 \text{ and } v = 0, \\ +\infty & \text{else.} \end{cases}$$

To shorten notation, we introduce the convex functions

$$F^{\alpha_f}(\rho_t, J_t, f_t) := \Psi(\rho_t, J_t) + \alpha_f \Psi(\rho_t, f_t)$$

as well as

$$\bar{F}^{\alpha_f}(\rho_t, J_t, f_t) = F^{\alpha_f}(\rho_t, J_t, f_t) + \frac{\alpha_{\text{inv}}}{2} |\rho_t - \rho_{\text{data}}|^2.$$

Then, it holds that

$$F^{\alpha_f}(\rho_t, J_t, f_t) = \sup_{\rho_t^*, J_t^*, f_t^*} \langle (\rho_t, J_t, f_t), (\rho_t^*, J_t^*, f_t^*) \rangle - (F^{\alpha_f})^*(\rho_t^*, J_t^*, f_t^*), \quad (3)$$

$$\bar{F}^{\alpha_f}(\rho_t, J_t, f_t) = \sup_{\rho_t^*, J_t^*, f_t^*} \langle (\rho_t, J_t, f_t), (\rho_t^*, J_t^*, f_t^*) \rangle - (\bar{F}^{\alpha_f})^*(\rho_t^*, J_t^*, f_t^*). \quad (4)$$

Substituting these expressions into our objective functional leads to the formulation

$$\begin{aligned} \sup_{(f_t, \rho_t, J_t)} \inf_{(\phi_t, \lambda_t), (\rho_t^*, J_t^*, f_t^*)} & \int_{((0,1) \times \Omega) \setminus \Gamma} (F^{\alpha_f})^*(\rho_t^*, J_t^*, f_t^*) dx dt + \int_{\Gamma} (\bar{F}^{\alpha_f})^*(\rho_t^*, J_t^*, f_t^*) dx dt \\ & + \mathcal{G}(\phi_t) + \langle (\partial_t \phi_t + \nabla \phi_t \cdot v - \rho_t^*, \nabla \phi_t - J_t^*, \phi_t - f_t^*), (\rho_t, J_t, f_t) \rangle_{\Omega} \\ & - \lambda_t \left(\int_{\Omega} d\rho_1 - \int_{\Omega} d\rho_0 + \int_0^1 \int_{\Omega} f_t dx dt \right), \end{aligned}$$

fitting the necessary structure for applying the classical method ALG2 from [?]. Moreover, calculating the convex conjugate of F^{α_f} gives

$$(F^{\alpha_f})^*(\rho_t^*, J_t^*, f_t^*) = \iota_{\{\rho_t^* + \frac{1}{2} \left(\frac{|J_t^*|^2}{1} + \frac{|f_t^*|^2}{\alpha_f} \right) \leq 0\}}$$

for the convex indicator function

$$\iota_C(x) := \begin{cases} 0 & \text{if } x \in C, \\ +\infty & \text{else.} \end{cases}$$

For $\bar{F}^{\alpha_f}$ we obtain (see calculations in Appendix)

$$(\bar{F}^{\alpha_f})^*(\rho_t^*, J_t^*, f_t^*) = \rho_{\text{dat}} \left(\rho_t^* + \frac{1}{2} (J_t^*)^2 + \frac{1}{2\alpha_f} (f_t^*)^2 \right) + \left(\frac{1}{\alpha_{\text{inv}}} - \frac{\alpha_{\text{inv}}}{2} \right) \left(\rho_t^* + \frac{1}{2} (J_t^*)^2 + \frac{1}{2\alpha_f} (f_t^*)^2 \right)^2$$

In this formulation, the variables ρ_t, J_t, f_t can be understood as Lagrange multipliers of the constraints $\partial_t \phi_t + \nabla \phi_t \cdot v - \rho_t^* = 0$, $\nabla \phi_t - J_t^* = 0$ and $\phi_t - f_t^* = 0$. We define the Augmented Lagrangian functional with parameter $r_1 > 0$ as

$$\begin{aligned} L(f_t, \rho_t, J_t, \phi_t, \lambda_t, \rho_t^*, J_t^*, f_t^*) &:= \int_{((0,1) \times \Omega) \setminus \Gamma} (F^{\alpha_f})^*(\rho_t^*, J_t^*, f_t^*) dxdt + \int_{\Gamma} (\overline{F}^{\alpha_f})^*(\rho_t^*, J_t^*, f_t^*) dxdt \\ &+ \mathcal{G}(\phi_t) + \langle (\partial_t \phi_t + \nabla \phi_t \cdot v_t - \rho_t^*, \nabla \phi_t - J_t^*, \phi_t - f_t^*), (\rho_t, J_t, f_t) \rangle_{\Omega} \\ &+ \frac{r_1}{2} \langle (\partial_t \phi_t + \nabla \phi_t \cdot v_t - \rho_t^*, \nabla \phi_t - J_t^*), (\partial_t \phi_t + \nabla \phi_t \cdot v_t - \rho_t^*, \nabla \phi_t - J_t^*) \rangle_{\Omega} \\ &+ \frac{r_1}{2} \langle \phi_t - f_t^*, \phi_t - f_t^* \rangle \\ &- \lambda_t \left(\int_{\Omega} d\rho_1 - \int_{\Omega} d\rho_0 + \int_0^1 \int_{\Omega} f_t dxdt \right), \end{aligned}$$

for $r_1 > 0$. The resulting problem reads

$$\sup_{(f_t, \rho_t, J_t)} \inf_{(\phi_t, \lambda_t), f_t^*, (\rho_t^*, J_t^*)} L(f_t, \rho_t, J_t, \phi_t, \lambda_t, \rho_t^*, J_t^*, f_t^*). \quad (5)$$

2.2 Algorithm

In this Section we present an algorithm solving the augmented lagrangian problem defined in (5). We will follow the structure of the method introduced in [?], modified to account for ...

The idea of the algorithm is to decouple the optimization of $(\phi_t), (\rho_t^*, J_t^*)$ and (ρ_t, J_t) . In what follows, $(\phi_m), (\rho_m^*, J_m^*)$ and (ρ_m, J_m) denote the approximate solutions after m iterations and, to improve readability, we omit the explicit time-dependence of the iterates. Each iteration consists of the following steps:

Step 1: Update (ϕ_t)

For this update, we aim at solving

$$\inf_{(\phi_{m+1})} L(f_t, \rho_t, J_t, \phi_t, \lambda_t, \rho_t^*, J_t^*, f_t^*)$$

In particular, the first variation with respect to ϕ_{m+1} needs to be zero as necessary condition, leading to the following equation

$$\begin{aligned} &r_1 \langle \partial_t \phi_{m+1}, \partial_t \psi_t \rangle_{\Omega} + r_1 \langle \nabla \phi_{m+1}, \nabla \psi_t \rangle_{\Omega} + r_1 \langle \phi_t, \psi_t \rangle_{\Omega} \\ &+ r_1 \langle \nabla \phi_t \cdot v, \nabla \psi_t \cdot v \rangle_{\Omega} + r_1 \langle \partial_t \phi_t, \nabla \psi_t \cdot v \rangle_{\Omega} + r_1 \langle \nabla \phi_t \cdot v, \partial_t \psi_t \rangle_{\Omega} \\ &= \langle \psi_t, \rho_t \rangle - \langle \psi_t, \rho_0 \rangle + \langle \partial_t \psi_t + \nabla \phi_t \cdot v, r_1 \rho_m^* - \rho_m \rangle_{\Omega} + \langle \nabla \psi_t, r_1 J_m^* - J_m \rangle_{\Omega} + \langle \psi_t, r_1 f_t^* - f_t \rangle_{\Omega} \end{aligned}$$

for test functions ψ_t . Introducing the bilinear-form

$$\begin{aligned} A(\phi_{m+1}, \psi_{m+1}) &:= r_1 \langle \partial_t \phi_{m+1}, \partial_t \psi_t \rangle_{\Omega} + r_1 \langle \nabla \phi_{m+1}, \nabla \psi_t \rangle_{\Omega} + r_1 \langle \phi_t, \psi_t \rangle_{\Omega} \\ &+ r_1 \langle \nabla \phi_t \cdot v, \nabla \psi_t \cdot v \rangle_{\Omega} + r_1 \langle \partial_t \phi_t, \nabla \psi_t \cdot v \rangle_{\Omega} + r_1 \langle \nabla \phi_t \cdot v, \partial_t \psi_t \rangle_{\Omega} \end{aligned}$$

and the linear form

$$F^m(\psi) := \langle \psi_t, \rho_t \rangle - \langle \psi_t, \rho_0 \rangle + \langle \partial_t \psi + \nabla \psi \cdot v, r_1 \rho_m^* - \rho_m \rangle + \langle \nabla \psi, r_1 J_m^* - J_m \rangle - \langle f_t - r_1 f_t^*, \psi_t \rangle_{\Omega}$$

the optimality system reads

$$A(\phi_{m+1}, \psi_t) = F^m(\psi_t).$$

Step 1b: Update (λ_t)

$$\lambda_{m+1} = \lambda_m - \left(\int_{\Omega} d\rho_1 - \int_{\Omega} d\rho_0 + \int_0^1 \int_{\Omega} f_t dxdt \right). \quad (6)$$

Step 2: Update (ρ_t^*, J_t^*, f_t^*) .

The new iterates $(\rho_{m+1}^*, J_{m+1}^*, f_{m+1}^*)$ are defined as the solution of (where we neglected terms constant in ρ^*, J^*, f^*)

$$\begin{aligned} & \inf_{(\rho_{m+1}^*, J_{m+1}^*, f_{m+1}^*)} \int_{((0,1) \times \Omega) \setminus \Gamma} (F^{\alpha_f})^*(\rho_t^*, J_t^*, f_t^*) \, dxdt + \int_{\Gamma} (\bar{F}^{\alpha_f})^*(\rho_t^*, J_t^*, f_t^*) \, dxdt \\ & - \langle \rho_{m+1}^*, \rho_m + r_1(\partial_t \phi_{m+1} + \nabla \phi_{m+1} \cdot v) \rangle_{\Omega} + \frac{r_1}{2} |\rho_{m+1}^*|_{\Omega}^2 \\ & - \langle J_{m+1}^*, J_m + r_1 \nabla \phi_{m+1} \rangle_{\Omega} + \frac{r_1}{2} |J_{m+1}^*|_{\Omega}^2 \\ & - \langle f_{m+1}^*, f_t + r_1 \phi_t \rangle + \frac{r_1}{2} |f_{m+1}^*|_{\Omega}^2 \end{aligned}$$

composed of two independent problems, one in $((0,1) \times \Omega) \setminus \Gamma$ behaving as in [?] and the other one on Γ . To simplify notation, we introduce the following variables containing the information from the previous iteration

$$\begin{aligned} \eta_{\rho^*} &:= \partial_t \phi_{m+1} + \nabla \phi_{m+1} \cdot v + \frac{1}{r_1} \rho_m, \\ \eta_{J^*} &:= \nabla \phi_{m+1} + \frac{1}{r_1} J_m, \\ \eta_{f^*} &:= \phi_{m+1} + \frac{1}{r_1} f_m \end{aligned}$$

and $\eta_{\Omega} := (\eta_{\rho^*}, \eta_{J^*}, \eta_{f^*})$, The minimization in $((0,1) \times \Omega) \setminus \Gamma$ problem reads as

$$(\rho_{m+1}^*, J_{m+1}^*, f_{m+1}^*) \stackrel{\iota}{\in} \left\{ \rho_{m+1}^* + \frac{1}{2} \left(\frac{|J_{m+1}^*|^2}{1} + \frac{|f_{m+1}^*|^2}{\alpha_f} \right) \leq 0 \right\} - r_1 \langle \eta_{\Omega}, (\rho_{m+1}^*, J_{m+1}^*, f_{m+1}^*) \rangle + \frac{r_1}{2} (|\rho_{m+1}^*|_{\Omega}^2 + |J_{m+1}^*|_{\Omega}^2 + |f_{m+1}^*|_{\Omega}^2)$$

If, η_{Ω} is admissible, then $(\rho_{m+1}^*, J_{m+1}^*) = \eta_{\Omega}$ is optimal. Otherwise let

$$\begin{aligned} x &:= |J_{m+1}^*| \\ y &:= f_{m+1}^*. \end{aligned}$$

Substituting these into the minimization and formulating the KKT system leads to

$$\begin{aligned} 0 &= x^3 + \frac{1}{\alpha_f} xy^2 + 2(1 + \eta_{\rho^*})x - 2\|\eta_{J^*}\| \\ 0 &= y^3 + \frac{\alpha_f}{1} x^2 y + 2(\alpha_f^2 + \alpha_f \eta_{\rho^*})y - 2\alpha_f^2 \eta_{f^*} \end{aligned}$$

instead. This system can be solved using Newton's method. On Γ , we have to solve the system

$$\begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} \rho_{\text{dat}} + 2 \left(\frac{1}{\alpha_{\text{inv}}} - \frac{\alpha_{\text{inv}}}{2} \right) \left(\rho_t^* + \frac{1}{2} (J_t^*)^2 + \frac{1}{2\alpha_f} (f_t^*)^2 \right) - r_1 \eta_{\rho^*} + r_1 \rho^* \\ \rho_{\text{dat}} J_t^* + J_t^* \left(\frac{1}{\alpha_{\text{inv}}} - \frac{\alpha_{\text{inv}}}{2} \right) \left(\rho_t^* + \frac{1}{2} (J_t^*)^2 + \frac{1}{2\alpha_f} (f_t^*)^2 \right) - r_1 \eta_{J^*} + r_1 J^* \\ \frac{1}{\alpha_f} \rho_{\text{dat}} f_t^* + \frac{f_t^*}{\alpha_f} \left(\frac{1}{\alpha_{\text{inv}}} - \frac{\alpha_{\text{inv}}}{2} \right) \left(\rho_t^* + \frac{1}{2} (J_t^*)^2 + \frac{1}{2\alpha_f} (f_t^*)^2 \right) - r_1 \eta_{f^*} + r_1 f^* \end{pmatrix}$$

Step 3: Update $(f_t, \rho_t, J_t; \rho_0, \rho_1)$

The new iterate will be defined as the solution of

$$\begin{aligned} & \sup_{(f_{m+1}, \rho_{m+1}, J_{m+1})} \langle (\partial_t \phi_{m+1} + \nabla \phi_{m+1} \cdot v - \rho_{m+1}^*, \nabla \phi_{m+1} - J_{m+1}^*, \phi_{m+1} - f_{m+1}^*), (\rho, J, f) \rangle_{\Omega} \\ & + \int_{\Omega} [\phi_1 d\rho_1 - \phi_0 d\rho_0] \\ & - \lambda_t \left(\int_{\Omega} d\rho_1 - \int_{\Omega} d\rho_0 + \int_0^1 \int_{\Omega} f_t \, dxdt \right), \end{aligned}$$

Algorithm 1 Augmented Lagrangian Method TO BE UPDATED AT THE END

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while  $m \leq it_{\max}^{ALG}$  and  $err_{ALG} > tol$  do
  Solve  $A_h(\phi_{m+1}) = F_h^m$  for  $\phi_{m+1}$ 
   $\eta_{\rho^*} \leftarrow \partial_t \phi_{m+1} + \frac{1}{r_1} \rho_m$  and  $\eta_{J^*} \leftarrow \nabla \phi_{m+1} + \frac{1}{r_1} J_m$ 
  if  $\eta_{\rho^*} + \frac{\eta_{J^*}^2}{2} \leq 0$  then
     $\rho_{m+1}^* \leftarrow \eta_{\rho^*}$  and  $J_{m+1}^* \leftarrow \eta_{J^*}$ 
  else
    Solve  $0 = x_{\Omega}^3 + 2(1 + \eta_{\rho^*})x_{\Omega} - 2\|\eta_{J^*}\|$ 
     $\rho_{m+1}^* \leftarrow -\frac{1}{2}x_{\Omega}^2$  and  $J_{m+1}^* \leftarrow x_{\Omega} \frac{\eta_{J^*}}{\|\eta_{J^*}\|}$ 
  end if
   $\rho_{m+1} \leftarrow \rho_m + r_1 (\partial_t \phi_{m+1} - \rho_{m+1}^* + v_t \cdot \nabla \phi_{m+1})$ 
   $J_{m+1} \leftarrow J_m + r_1 (\nabla \phi_{m+1} - J_{m+1}^*)$ 
   $err_{\Omega} \leftarrow \max(\|\partial_t \phi_{m+1} - \rho_{m+1}^* + v_t \cdot \nabla \phi_{m+1}\|_{\infty}, \|\nabla \phi_{m+1} - J_{m+1}^*\|_{\infty})$ 
   $err_{ALG} \leftarrow err_{\Omega}$ 
   $m \leftarrow m + 1$ 
end while

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and approximated using one gradient descent step given as

$$\begin{aligned}
 \rho_{m+1} &:= \rho_m + r_1 (\partial_t \phi_{m+1} + \nabla \phi_{m+1} \cdot v - \rho_{m+1}^*) \\
 J_{m+1} &:= J_m + r_1 (\nabla \phi_{m+1} - J_{m+1}^*) \\
 f_{m+1} &= f_m + r_1 (\phi_{m+1} - f_{m+1}^*) + \lambda_t \\
 \rho_{0,m+1} &= \rho_{0,m} - \phi_0 - \lambda_t, \\
 \rho_{1,m+1} &= \rho_{1,m} + \phi_1 + \lambda_t
 \end{aligned}$$

One iteration of these three steps is called one step of the Augmented Lagrangian method, ALG for short, and is summarized in Algorithm 1.

A Calculation of convex conjugate of $\overline{F}^{\alpha_f}$

Our starting point is

$$(\overline{F}^{\alpha_f})^*(\rho_t^*, J_t^*, f_t^*) = \sup_{(\rho_t, J_t, f_t)} \rho_t \rho_t^* + J_t J_t^* + f_t f_t^* - \frac{|J_t|^2}{2\rho_t} - \alpha_f \frac{|f_t|^2}{2\rho_t} - \frac{\alpha_{\text{inv}}}{2} |\rho_t - \rho_{\text{dat}}|^2 \quad (7)$$

The optimality conditions read as

$$\begin{aligned}
 0 &= \rho_t^* + \frac{|J_t|^2}{2\rho_t^2} + \alpha_f \frac{|f_t|^2}{2\rho_t^2} - \alpha_{\text{inv}}(\rho_t - \rho_{\text{dat}}) \\
 0 &= J_t^* - \frac{J_t}{\rho_t} \\
 0 &= f_t^* - \alpha_f \frac{f_t}{\rho_t}
 \end{aligned}$$

Using the two last equations to replace J_t and f_t in (7) yields

$$\sup_{\rho_t} \rho_t \left(\rho_t^* + \frac{1}{2} (J_t^*)^2 + \frac{1}{2\alpha_f} (f_t^*)^2 \right) - \frac{\alpha_{\text{inv}}}{2} |\rho_t - \rho_{\text{dat}}|^2 \quad (8)$$

Next, we calculate the optimality of this remaining problem

$$0 = \rho_t^* + \frac{1}{2} (J_t^*)^2 + \frac{1}{2\alpha_f} (f_t^*)^2 - \alpha_{\text{inv}} \rho_t + \alpha_{\text{inv}} \rho_{\text{dat}}$$

i.e.

$$\rho_t = \rho_{\text{dat}} + \frac{1}{\alpha_{\text{inv}}} \left(\rho_t^* + \frac{1}{2}(J_t^*)^2 + \frac{1}{2\alpha_f}(f_t^*)^2 \right)$$

Inserting this expression back into (8), we obtain

$$(\overline{F}^{\alpha_f})^*(\rho_t^*, J_t^*, f_t^*) = \rho_{\text{dat}} \left(\rho_t^* + \frac{1}{2}(J_t^*)^2 + \frac{1}{2\alpha_f}(f_t^*)^2 \right) + \left(\frac{1}{\alpha_{\text{inv}}} - \frac{\alpha_{\text{inv}}}{2} \right) \left(\rho_t^* + \frac{1}{2}(J_t^*)^2 + \frac{1}{2\alpha_f}(f_t^*)^2 \right)^2.$$

To solve for new dual variable, we have to minimize the functional

$$(\overline{J}^{\alpha_f})^*(\rho_t^*, J_t^*, f_t^*) := (\overline{F}^{\alpha_f})^*(\rho_t^*, J_t^*, f_t^*) - r_1 \langle \eta_{\Omega}, (\rho_{m+1}^*, J_{m+1}^*, f_{m+1}^*) \rangle + \frac{r_1}{2} (|\rho_{m+1}^*|_{\Omega}^2 + |J_{m+1}^*|_{\Omega}^2 + |f_{m+1}^*|_{\Omega}^2)$$

with gradient

$$\nabla_{\rho_t^*, J_t^*, f_t^*} (\overline{J}^{\alpha_f})^*(\rho_t^*, J_t^*, f_t^*) = \begin{pmatrix} \rho_{\text{dat}} + 2 \left(\frac{1}{\alpha_{\text{inv}}} - \frac{\alpha_{\text{inv}}}{2} \right) \left(\rho_t^* + \frac{1}{2}(J_t^*)^2 + \frac{1}{2\alpha_f}(f_t^*)^2 \right) - r_1 \eta_{\rho^*} + r_1 \rho^* \\ \rho_{\text{dat}} J_t^* + 2 J_t^* \left(\frac{1}{\alpha_{\text{inv}}} - \frac{\alpha_{\text{inv}}}{2} \right) \left(\rho_t^* + \frac{1}{2}(J_t^*)^2 + \frac{1}{2\alpha_f}(f_t^*)^2 \right) - r_1 \eta_{J_t^*} + r_1 J_t^* \\ \frac{1}{\alpha_f} \rho_{\text{dat}} f_t^* + 2 \frac{f_t^*}{\alpha_f} \left(\frac{1}{\alpha_{\text{inv}}} - \frac{\alpha_{\text{inv}}}{2} \right) \left(\rho_t^* + \frac{1}{2}(J_t^*)^2 + \frac{1}{2\alpha_f}(f_t^*)^2 \right) - r_1 \eta_{f^*} + r_1 f^* \end{pmatrix}$$

More explicitly

$$\nabla_{\rho_t^*, J_{t,x}^*, J_{t,y}^*, f_t^*} (\overline{J}^{\alpha_f})^*(\rho_t^*, J_t^*, f_t^*) = \begin{pmatrix} \rho_{\text{dat}} + 2 \left(\frac{1}{\alpha_{\text{inv}}} - \frac{\alpha_{\text{inv}}}{2} \right) \left(\rho_t^* + \frac{1}{2}(J_t^*)^2 + \frac{1}{2\alpha_f}(f_t^*)^2 \right) - r_1 \eta_{\rho^*} + r_1 \rho^* \\ \rho_{\text{dat}} J_{t,x}^* + 2 J_{t,x}^* \left(\frac{1}{\alpha_{\text{inv}}} - \frac{\alpha_{\text{inv}}}{2} \right) \left(\rho_t^* + \frac{1}{2}(J_t^*)^2 + \frac{1}{2\alpha_f}(f_t^*)^2 \right) - r_1 \eta_{J_{t,x}^*} + r_1 J_{t,x}^* \\ \rho_{\text{dat}} J_{t,y}^* + 2 J_{t,y}^* \left(\frac{1}{\alpha_{\text{inv}}} - \frac{\alpha_{\text{inv}}}{2} \right) \left(\rho_t^* + \frac{1}{2}(J_t^*)^2 + \frac{1}{2\alpha_f}(f_t^*)^2 \right) - r_1 \eta_{J_{t,y}^*} + r_1 J_{t,y}^* \\ \frac{1}{\alpha_f} \rho_{\text{dat}} f_t^* + 2 \frac{f_t^*}{\alpha_f} \left(\frac{1}{\alpha_{\text{inv}}} - \frac{\alpha_{\text{inv}}}{2} \right) \left(\rho_t^* + \frac{1}{2}(J_t^*)^2 + \frac{1}{2\alpha_f}(f_t^*)^2 \right) - r_1 \eta_{f^*} + r_1 f^* \end{pmatrix}$$

With $\bar{\alpha}_{\text{inv}} = 2 \left(\frac{1}{\alpha_{\text{inv}}} - \frac{\alpha_{\text{inv}}}{2} \right)$ and $K = \left(\rho_t^* + \frac{1}{2}(J_t^*)^2 + \frac{1}{2\alpha_f}(f_t^*)^2 \right)$, Jacobian matrix reads as

$$\begin{pmatrix} 2\bar{\alpha}_{\text{inv}} + r_1 & 2\bar{\alpha}_{\text{inv}}(J_{t,x}^* + J_{t,y}^*) & 2\bar{\alpha}_{\text{inv}}(J_{t,x}^* + J_{t,y}^*) & \frac{2}{\alpha_f} \bar{\alpha}_{\text{inv}} f_t^* \\ J_{t,x}^* \bar{\alpha}_{\text{inv}} & \rho_{\text{dat}} + \bar{\alpha}_{\text{inv}} K + \bar{\alpha}_{\text{inv}} J_{t,x}(J_{t,x}^* + J_{t,y}^*) + r_1 & \bar{\alpha}_{\text{inv}} J_{t,x}(J_{t,x}^* + J_{t,y}^*) & \frac{\bar{\alpha}_{\text{inv}}}{\alpha_f} J_{t,x}^* f_t^* \\ J_{t,y}^* \bar{\alpha}_{\text{inv}} & \bar{\alpha}_{\text{inv}} J_{t,y}(J_{t,x}^* + J_{t,y}^*) & \rho_{\text{dat}} + \bar{\alpha}_{\text{inv}} K + \bar{\alpha}_{\text{inv}} J_{t,y}(J_{t,x}^* + J_{t,y}^*) + r_1 & \frac{\bar{\alpha}_{\text{inv}}}{\alpha_f} J_{t,y}^* f_t^* \\ \frac{f_t^*}{\alpha_f} \bar{\alpha}_{\text{inv}} & \frac{f_t^*}{\alpha_f} \bar{\alpha}_{\text{inv}}(J_{t,x}^* + J_{t,y}^*) & \frac{f_t^*}{\alpha_f} \bar{\alpha}_{\text{inv}}(J_{t,x}^* + J_{t,y}^*) & \frac{1}{\alpha_f} \rho_{\text{dat}} + \frac{1}{\alpha_f} \bar{\alpha}_{\text{inv}} K + \frac{(f_t^*)^2}{(\alpha_f)^2} \bar{\alpha}_{\text{inv}} + r_1 \end{pmatrix}$$

References